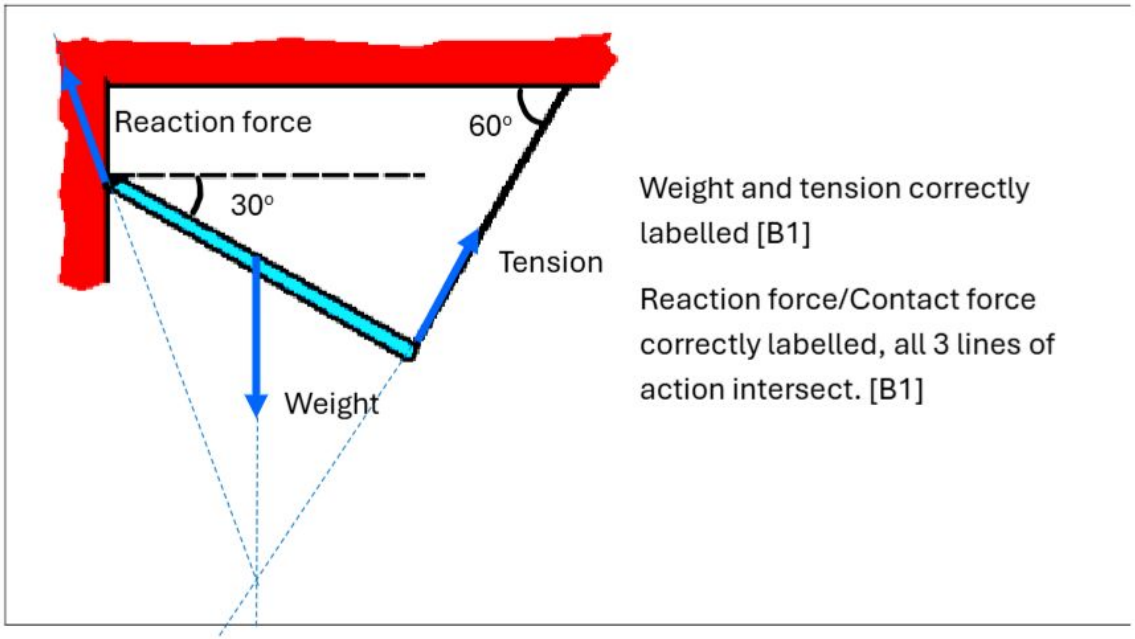
1 (a) The moment of a force is defined as the product of the force and the perpendicular distance from the pivot to the line of action of the force. B1

(b)(i)



(ii) Take moments about the hinge,

$30 \times 9.81 \times \cos 30° (\frac{L}{2}) = T \times L$ M1

$T = 127.4 \approx 127 N$ A0

(iii) M1

$R_X = T\cos 60^o = 127.4\cos 60^{oo} = 63.7 N$

$R_Y + T_Y = W$

$R_Y + 127 \sin 60° = 30(9.81)$

$R_Y = 184$ N M1

$R = \sqrt{R_X^2 + R_Y^2} = \sqrt{63.7^2 + 184^2} = 194.7 \approx 195 N$ A1

$\tan\theta = \frac{184}{63.7} \Rightarrow \theta = 70.9°$ above the horizontal A1

2 (a) The total momentum of a system remains constant if no resultant external force acts on the